A Comprehensive Review of Machine Learning Techniques for Intelligent Personal Finance Management Systems

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Abstract

The swift rise of fintech platforms and smart Personal Finance Management Systems (PFMS) has significantly changed the way individuals and groups handle their income, expenses, and financial choices. Although there has been notable advancement, current research is still scattered across various segments, such as budgeting, expense forecasting, anomaly detection, explainable financial analytics, and group expense management. This fragmentation limits the creation of cohesive and intelligent PFMS solutions. This review delves into the essential elements of contemporary PFMS, which encompass expense tracking, bill splitting, predictive budgeting, financial anomaly detection, recommendation systems, and explainable AI methods. The analysis covers a wide range of approaches, including statistical methods, machine learning, deep learning, and hybrid techniques, all tailored for personal and collaborative financial scenarios. By employing a qualitative literature survey and comparative analysis, this review evaluates the trends, strengths, and weaknesses of the existing systems. It offers a well-organized taxonomy of PFMS components and provides comparative insights across various learning methods. Additionally, it pinpoints gaps in current research to steer future efforts toward developing integrated, explainable, and user-friendly intelligent finance systems.

Keywords: Personal Finance Management Systems, Machine Learning, Expense Tracking, Budgeting and Forecasting, Group Expense Management, Anomaly Detection, Explainable Artificial Intelligence

1 Introduction

In today's digital age, managing personal and group finances has become a more intricate task. With online payments, mobile wallets, subscription services, and the shift towards cashless transactions, individuals find themselves navigating beyond basic income and expense tracking. They now face the challenge of managing fluctuating spending habits, shared costs, budgeting limitations, and planning for future financial needs.

This shift has led to an increased reliance on Personal Finance Management Systems (PFMS) and applications designed for group expense management. These tools are critical in helping users keep tabs on, analyze, and control their financial activities. However, research indicates that many existing systems lack the necessary intelligence and adaptability. Most traditional PFMS serve primarily as digital ledgers, depending on rigid rules, manual sorting, and basic summaries.

While these tools can assist with fundamental budgeting and expense tracking, they often fall short in personalization and predictive insights. Likewise, group expense management apps, like those for splitting bills, tend to concentrate on calculating balances and settlements without providing deeper analytical insights, spotting anomalies, or offering smart recommendations.

These shortcomings have spurred researchers to look into machine learning techniques to improve budgeting, forecasting, anomaly detection, and collaborative finance management. Recent studies highlight the use of various machine learning and statistical models across different elements of PFMS. For instance, budgeting and expense analysis often utilize methods such as Exponentially Weighted Moving Averages (EWMA), clustering techniques, Random Forests, ARIMA, and Long Short-Term Memory (LSTM) models to reveal spending patterns and manage budget constraints.

Research focused on forecasting applies similar models to scrutinize past financial data and project future expenses. When it comes to detecting financial anomalies and providing AI-driven advice, unsupervised and semi-supervised methods like Isolation Forest are explored to spot unusual or risky spending behaviors. In the realm of group expense management and bill splitting, regression-based models—including linear regression and other predictive approaches—are examined to understand shared spending trends and settlement patterns.

Despite these advancements, the existing literature tends to be scattered. Most studies address individual components in isolation, with budgeting, forecasting, anomaly detection, AI recommendations, and group finance rarely examined together within a cohesive PFMS framework. Furthermore, there's a lack of comparative analysis across models in these areas, which makes it challenging to grasp their respective strengths, weaknesses, and suitability for intelligent personal finance systems.

This review seeks to bridge that gap by systematically exploring and contrasting machine learning models across various PFMS components, including budgeting, forecasting, anomaly detection, AI-driven recommendations, and group expense management. By bringing together insights from available research, this review aspires to offer a clearer understanding of how different models are implemented, compared, and assessed within intelligent PFMS, thereby shedding light on current practices and ongoing research challenges.

2 Overview of Personal Finance Management Systems

2.1 Existing System

The existing personal finance management system, also known as PFMS, as discussed in the literature above, operates on the basis of using deterministic and rule-based driven mechanisms. All the financial transactions provided by the users are first processed through the predefined logic for expense based categorisation, aggregation, prediction and budget based validation. An important factor when it comes to Budgeting in general, are the thresholds that are established where alerts are generated once the limit exceeds. Detailed summaries are also provided over the duration of fixed time windows such as daily or monthly periods.

With respect to group expense management systems, overall balances are calculated using static rules like equal splits or user spending based splitting, followed by manual or semi-automated settlement tracking. These systems combined, provide a basic monitoring and data transparency feature, while assuming that the user's spending behaviour and inputs are always accurate. This results in the PFMS system becoming reactive in nature, rather than concluding deterministic outcomes itself- which makes it's adaptability very limited and contributes to the inability of recognising patterns in data that are crucial for personalisation and predicitive insights.

2.2 Proposed Intelligent PFMS

To overcome issues like adaptability, incompliance and inability to work around non-linear data, recent research introduces the "intelligent PFMS paradigms" that extends beyond the capabilities of traditional PFMS pipelines, with adaptive and data-driven components. Instead of relying on rule-based systems, proposed system has integrated learning based mechanisms for features like budgeting, forecasting, anomaly detection, AI-driven suggestions and group expense analysis pipelines.

These systems focus on analysing historical financial data to identify behavioural patterns, support adaptive budget management, and estimate future expenses while also detecting any irregular or unexpected spending activities. In collaborative finance scenarios, intelligent PFMS enhances balance analysis and settlement reasoning by incorporating behavioural trends instead of simply using static splits alone.

The key differentiator for the proposed system lies within its ability to evolve along with the user data, this allows PFMS to shift from descriptive reporting to decision support. While these systems improve automation and contextual awareness, the literature also highlights unresolved challenges relative to explainability, data quality, privacy and user trust issues. This indicates that intelligence must be introduced incrementally rather than a complete replacement of existing structures.

2.3 Mathematical Model

At the system level, PFMS trends can be represented using a common mathematical abstraction which is suitable for both existing and intelligent paradigms. Let the financial transaction sequence be defined as:

$$X = \{x_1, x_2, \dots, x_n\} \tag{1}$$

where each x_i represents an individual financial transaction.

In existing PFMS, system outputs such as expense summaries, budget alerts, or group balances are generated using deterministic mappings:

$$Y = g(X) \tag{2}$$

where $g(\cdot)$ denotes fixed aggregation rules and threshold-based constraints.

In contrast, proposed intelligent PFMS model the financial system as an adaptive mapping:

$$\hat{Y} = f(X \mid \Theta) \tag{3}$$

where Θ represents learned financial behavior patterns used to support budgeting decisions, forecasting, anomaly identification, and group expense analysis. Despite this difference, both paradigms follow the same high-level flow:

Input Data → Financial Processing → Decision-Support Output

This abstraction enables a clear comparison between static and intelligent PFMS and provides a consistent foundation for system-level analysis and diagrammatic representation.

3 Budgeting Techniques in Personal Finance

Budgeting within Personal Finance Management Systems (PFMS) is fundamentally a time-dependent activity, as individual and group spending behavior changes continuously across different time horizons. Unlike static financial summaries, effective budgeting requires understanding how expenses evolve over time in response to income variations, lifestyle changes, and external factors. Prior research emphasizes that budgeting mechanisms must account for historical trends, short-term fluctuations, and long-term behavioral patterns.

Consequently, PFMS budgeting approaches have progressed from static rule enforcement to statistically driven, behavior-aware, and temporally modeled techniques.

3.1 Rule-based budgeting

Rule-based budgeting constitutes the most basic form of budget control adopted in PFMS. In this approach, expenses accumulated within a predefined time period are summed and evaluated against a fixed budget threshold. Let the sequence of expenses within a budgeting interval be denoted as

$$X = \{x_1, x_2, \dots, x_n\} \tag{4}$$

The total expenditure is calculated as

$$E = \sum_{i=1}^{n} x_i \tag{5}$$

and budget compliance is assessed using a deterministic condition:

$$\text{Alert} = \begin{cases} 1, & \text{if } E > B \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

Such mechanisms are widely adopted due to their clarity and ease of interpretation. However, since budget limits remain constant over time, this approach implicitly assumes consistent spending behavior and offers limited adaptability when financial patterns change.

3.2 Exponentially Weighted Moving Averages (EWMA)

To introduce temporal responsiveness, statistical budgeting techniques have been explored in PFMS literature. Approaches based on Exponentially Weighted Moving Averages (EWMA) incorporate past expenditure while assigning greater influence to recent spending activity. The smoothed expenditure estimate is expressed as

$$S_t = \alpha x_t + (1 - \alpha) S_{t-1} \tag{7}$$

where x_t represents current expenditure and α controls the weight assigned to recent observations. EWMA-based budgeting enables gradual adjustment of budget baselines, making it suitable for monitoring short-term spending trends. Nonetheless, these methods presume relatively smooth behavioral transitions and may be affected by sudden or irregular expense patterns.

3.3 Behavior-oriented budgeting

Behavior-oriented budgeting approaches further extend temporal analysis through clustering techniques that group expenses or users according to similarity in spending behavior over time. Given a set of financial observations X, clustering partitions the data into multiple behavioral groups:

$$X = \bigcup_{k=1}^{K} C_k \tag{8}$$

Budget estimates are derived from cluster-level characteristics rather than uniform thresholds. This enables PFMS to reflect heterogeneous spending habits across users. However, the effectiveness of such approaches is closely tied to data quality and historical availability, and the resulting behavioral groupings may not always be easily interpretable.

3.4 Long Short-Term Memory (LSTM)

Deep learning-based budgeting approaches further generalize time-series modeling by capturing long-term dependencies in expenditure sequences using Long Short-Term Memory (LSTM) networks. LSTM-based formulations regulate information flow through input, forget, and output mechanisms to retain relevant financial context over extended time horizons.

Future expenditure is estimated as a function of historical observations:

$$x_{t+1} = f(X_t) \tag{9}$$

and the corresponding budget is computed using a buffered prediction:

$$B_{t+1} = x_{t+1} + \delta \tag{10}$$

These approaches support adaptive budgeting aligned with evolving financial behavior, while introducing challenges related to data dependency and interpretability.

Overall, the reviewed literature indicates a transition from static budget enforcement toward adaptive and predictive budgeting formulations, reflecting the increasing temporal complexity of personal financial data.

The figure depicts a comparison between the actual time-series values and the corresponding predicted values generated during the training phase. The observed series exhibits high short-term volatility with frequent local fluctuations, reflecting the irregular nature of financial data. In contrast, the predicted curve follows a smoother trajectory, capturing the underlying trend while suppressing noise. This behavior indicates that the model emphasizes long-term temporal patterns rather than reacting to transient variations. Such smoothing is desirable in personal finance forecasting, where trend awareness is more critical than exact point-wise replication of short-term fluctuations.

3.4.1 Internal Structure and Functional Mechanism of Long Short-Term Memory (LSTM)

Long Short-Term Memory (LSTM) networks are described in the literature as advanced recurrent architectures developed to address the challenge of retaining information across long temporal sequences. Conventional recurrent models often struggle to preserve contextual information over extended intervals, particularly when modeling sequential data such as financial transactions. LSTM overcomes this limitation by incorporating a dedicated memory component that enables selective storage and controlled modification of information over time.

The internal functioning of an LSTM unit is regulated by three interconnected gating mechanisms—namely the forget gate, input gate, and output gate—which interact with a shared

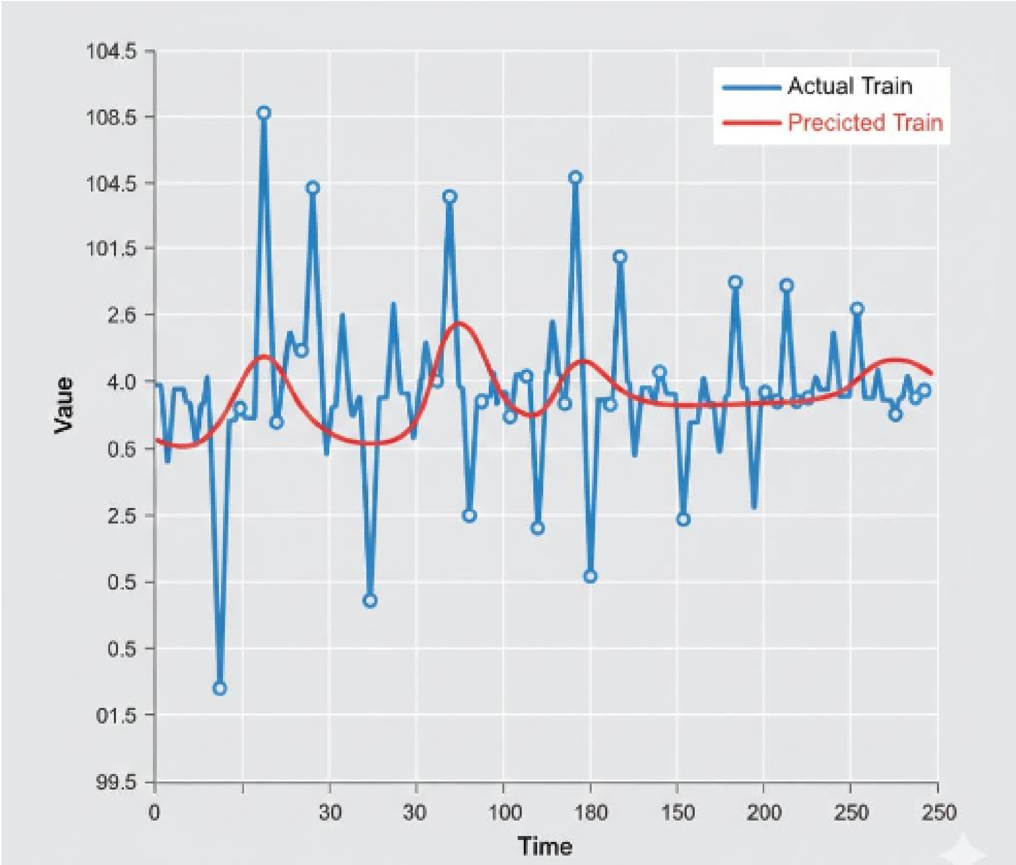

Figure 1: Actual vs Predicted Values using LSTM

internal memory cell. These gates collectively manage the flow of information through the network and determine how historical and current data are represented.

The forget gate evaluates the relevance of previously stored information and determines which portions of the historical context should be retained or discarded. In financial modeling scenarios, this mechanism allows the network to remove outdated spending behaviors while maintaining persistent patterns that remain significant over time.

$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right) \tag{11}$$

The input gate governs the integration of new information into the internal memory. It assesses the importance of recent financial activity and controls how much of this information is added to the memory state. This selective updating process ensures that new spending trends are incorporated without disrupting meaningful long-term financial representations.

$$i_t = \sigma\left(W_i \cdot [h_{t-1}, x_t] + b_i\right) \tag{12}$$

The internal memory cell serves as a continuous state that accumulates and preserves relevant information across time steps. By combining retained historical patterns with newly admitted data, the memory cell enables the network to maintain temporal continuity in sequential representations.

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tanh\left(W_C \cdot [h_{t-1}, x_t] + b_C\right) \tag{13}$$

The output gate regulates the extent to which the internal memory influences the current output. It ensures that only contextually relevant information is propagated forward, thereby shaping subsequent estimates or analytical outcomes.

LSTM architecture is highlighted for its capacity to capture evolving spending behavior while preserving long-term financial habits. At the same time, the literature acknowledges challenges related to transparency, interpretability, and computational demands, underscoring the importance of integrating LSTM-based approaches thoughtfully within broader Personal Finance Management System frameworks.

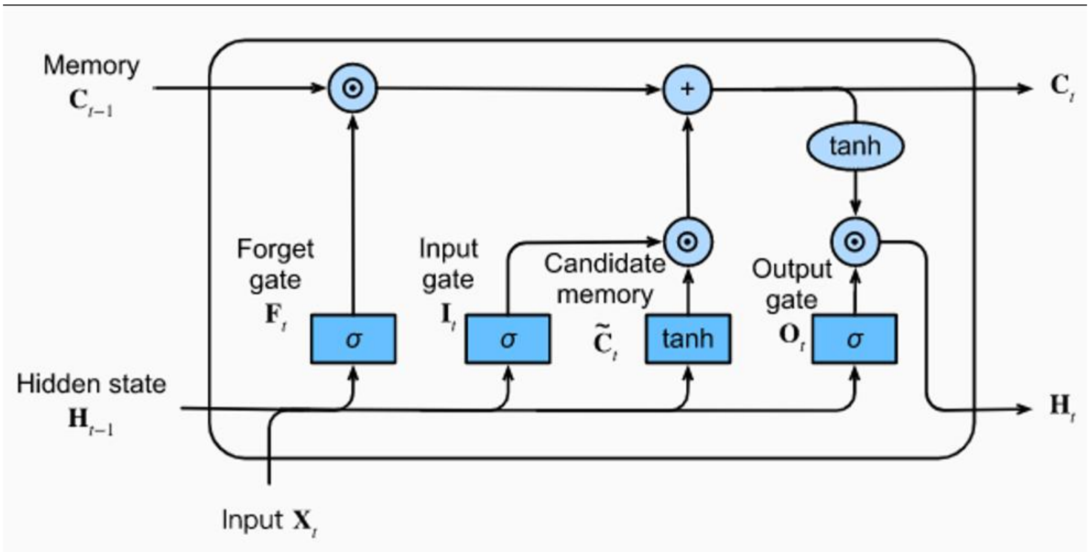

Figure 2: LSTM architecture

Building on the system-level workflow, the discussion now focuses on the budgeting pipeline within Personal Finance Management Systems. This pipeline is examined in terms of how expenditure data is processed over time to support budget formulation and monitoring. The subsequent sections review the techniques commonly employed at different stages of the budgeting process, highlighting their role in handling temporal spending behavior and uncertainty.

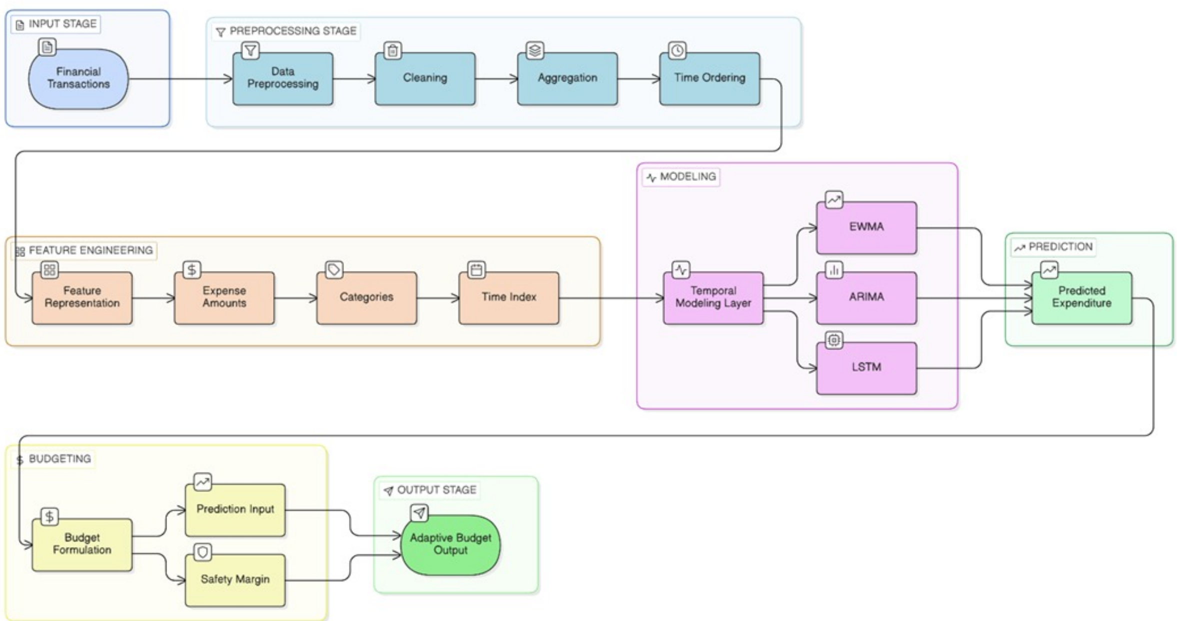

Figure 3: Conceptual Architecture of the Budgeting Pipeline

This Figure illustrates a generalized budgeting pipeline in Personal Finance Management

Systems. Financial transaction data first enters the input stage and undergoes preprocessing operations, including cleaning, aggregation, and temporal ordering, to ensure data consistency. The processed data is then transformed through feature engineering, where expenditure values, categorical information, and time indices are represented for analysis. A temporal modeling layer applies statistical and learning-based techniques to capture expenditure dynamics and generate predicted expenditure estimates. These predictions are forwarded to the budgeting stage, where a safety margin is incorporated during budget formulation to account for uncertainty. The final output is an adaptive budget representation that evolves with observed spending behavior. The architecture represents a conceptual workflow synthesized from existing literature rather than a specific system implementation.

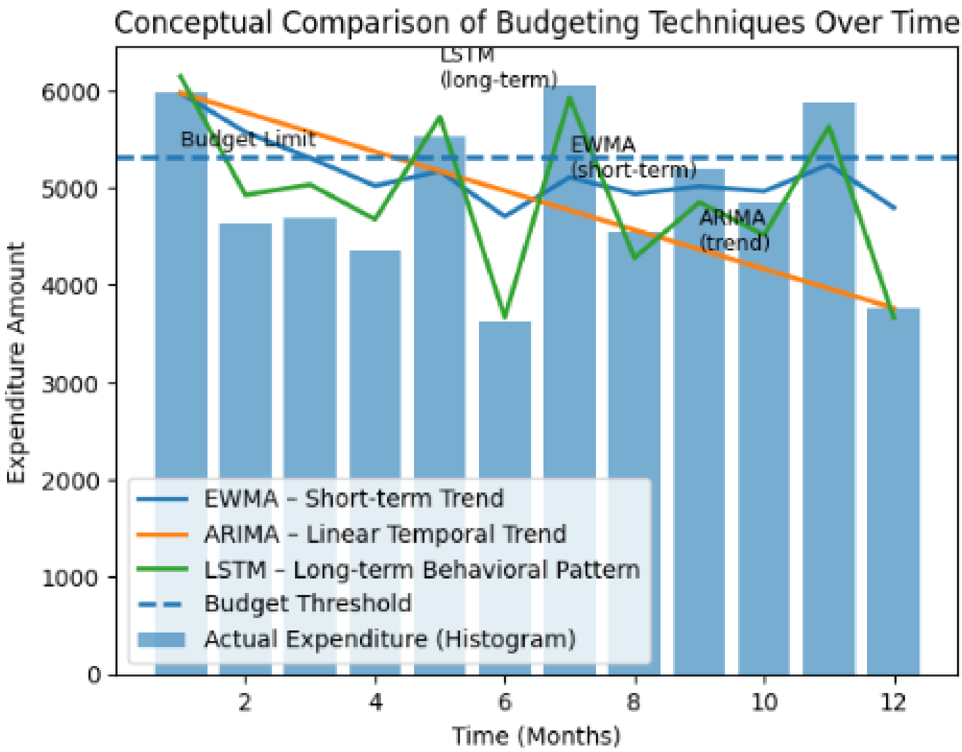

Figure 4: Visualization of budgeting techniques in personal finance management

The diagram illustrates how multiple budgeting techniques analyze the same expenditure data across time. Actual spending is represented using a histogram, highlighting natural variability in financial behavior. The EWMA curve demonstrates short-term smoothing by emphasizing recent expenditures, while the ARIMA trend reflects linear temporal dependencies observed in historical data. The LSTM-based curve captures longer-term and non-linear spending patterns, representing behavioral dynamics that evolve over extended periods. The budget threshold serves as a reference boundary for evaluating budget compliance. Overall, the diagram conveys how different budgeting approaches provide complementary perspectives on time-dependent financial behavior rather than competing solutions. Having examined budgeting techniques and their temporal behavior, the discussion now shifts to forecasting mechanisms employed in Personal Finance Management Systems.

4 Forecasting Techniques in Personal Finance

Forecasting represents a fundamental analytical function in Personal Finance Management Systems (PFMS), enabling a shift from retrospective financial reporting to proactive financial decision support. By estimating future expenditure trajectories, forecasting mechanisms assist users in identifying potential budgetary constraints, assessing savings progression, and adapting spending behavior before financial instability occurs. Beyond aggregate predictions, category-level forecasting—covering domains such as food, transportation, utilities, and discretionary expenses—supports structured resource allocation and informed long-term planning. Consequently, forecasting plays a central role in enhancing the intelligence and practical utility of modern PFMS.

4.1 Statistical Time-Series Models (ARIMA and SARIMA)

Statistical time-series models have traditionally served as the foundation of financial forecasting due to their mathematical transparency and computational efficiency. Among these, the AutoRegressive Integrated Moving Average (ARIMA) model is widely employed to capture linear temporal dependencies in expenditure data. ARIMA estimates future values as a function of historical observations and past forecast errors by combining autoregressive, differencing, and moving average components. The general formulation of an ARIMA model is expressed as

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \cdots + \theta_q \epsilon_{t-q} + \epsilon_t \tag{14}$$

where Y_t denotes the stationary time-series value at time t, ϕ represents autoregressive coefficients, and θ denotes moving average parameters. To account for periodic spending behavior commonly observed in consumer finance, Seasonal ARIMA (SARIMA) extends this formulation by incorporating seasonal components. These statistical models are typically applied in short-term forecasting scenarios where expenditure trends are relatively stable and stationarity assumptions are satisfied.

Figure 5: Comparison of ARMA, ARIMA, and SARIMA time-series models

4.2 Machine Learning Regression Approaches

Machine learning regression methods provide an alternative to classical statistical models by enabling the capture of non-linear relationships within high-dimensional financial data. In PFMS literature, ensemble-based regression techniques are frequently examined due to their robustness and ability to model heterogeneous spending patterns. The effectiveness of these approaches depends substantially on feature engineering processes that transform raw transaction logs into structured inputs. Common representations include rolling expenditure averages over fixed temporal windows, trend indicators that reflect spending increases or declines, and category-specific frequency measures. Such features allow regression models to learn complex financial interactions that are difficult to represent using linear time-series formulations.

4.3 Deep Learning Methods (LSTM and GRU)

Deep learning architectures are increasingly explored for forecasting tasks involving complex and long-term temporal dependencies. Recurrent neural models, particularly Long Short-Term Memory (LSTM) networks, are designed to preserve relevant historical information while adapting to recent expenditure variations through gated memory mechanisms. This architectural design enables the modeling of recurring expenses, seasonal patterns, and evolving financial behavior across extended time horizons.

Gated Recurrent Units (GRU) are also discussed as a computationally efficient alternative, employing a simplified gating structure while retaining the capacity to model sequential dependencies. Such architectures are often considered in scenarios where computational constraints or real-time processing requirements limit the applicability of more complex recurrent models.

4.4 Hybrid and Ensemble Forecasting Approaches

To address the limitations associated with standalone forecasting models, recent research investigates hybrid and ensemble forecasting frameworks that integrate multiple modeling paradigms. A commonly reported approach combines ARIMA and LSTM outputs, leveraging the strength of ARIMA in capturing linear temporal trends and the capability of LSTM to model non-linear residual behavior. In these ensemble formulations, individual model predictions are aggregated using weighted combinations:

$$Y_{t,\text{ensemble}} = w_1 \cdot Y_{t,\text{ARIMA}} + w_2 \cdot h_{t,\text{LSTM}} \tag{15}$$

where w_1 and w_2 represent weighting coefficients optimized through validation procedures. Such hybrid strategies aim to enhance forecast robustness across varying financial conditions without relying exclusively on a single modeling assumption.

4.5 Cross-Cutting Discussion

Across forecasting methodologies, several cross-cutting considerations consistently emerge in the literature. Forecast horizon significantly influences model suitability, with statistical approaches commonly applied to short-term prediction tasks, while deep learning models are examined for longer-term forecasting where seasonal and behavioral patterns extend over months or years. Data availability and quality also play a critical role, as deep learning techniques typically require substantial historical data and are sensitive to noise, whereas statistical models are more resilient in data-sparse settings but require stationarity through preprocessing techniques.

Interpretability remains a central concern in PFMS applications. Statistical models are often regarded as transparent systems, while deep learning approaches introduce opacity that motivates the integration of explainable artificial intelligence mechanisms to support user trust and regulatory compliance. Additionally, scalability and deployment constraints are highlighted, particularly for mobile and web-based PFMS, where computationally intensive models must be managed through layered system architectures and API-driven services to ensure efficient operation.

5 AI-Powered Spending Insights and Anomaly Detection

5.1 Distinction Between Anomaly Detection and Fraud Detection

In Personal Finance Management Systems (PFMS), anomaly detection and fraud detection represent distinct analytical objectives, despite being frequently conflated in financial literature. Fraud detection is conventionally formulated as a supervised learning problem, relying on historical datasets annotated with explicit fraud labels to train classification models. Such formulations are well suited to institutional banking environments where large, labeled datasets are available.

In contrast, spending anomaly detection within PFMS is inherently an unsupervised task. Its primary objective is to identify atypical, inefficient, or irregular spending behavior relative to an individual user's historical patterns, without requiring predefined labels of undesirable activity. This distinction arises from the highly personalized nature of spending behavior and the practical difficulty of generating labeled datasets that define overspending or abnormal consumption for individual users. As a result, anomaly detection in PFMS focuses on deviation analysis rather than categorical identification of fraudulent events.

5.2 Rule-Based and Statistical Detection Methods

Early PFMS implementations employ rule-based logic and statistical thresholds as baseline mechanisms for identifying anomalous spending behavior. These methods typically assume a stable distribution of expenditure data and flag transactions that exceed expected variability limits. A common statistical formulation defines an anomaly when current expenditure exceeds a rolling historical average by a multiple of the observed dispersion:

$$\text{Anomaly} = \text{True if } x_{\text{current}} > \mu_{\text{window}} + k \cdot \sigma_{\text{window}} \tag{16}$$

where μ_{window} and σ_{window} denote the mean and standard deviation computed over a recent temporal window, and k is a predefined scaling factor. While such approaches are computationally efficient and easily interpretable, their effectiveness is limited in PFMS contexts. They are unable to account for multi-feature spending behavior or adapt to seasonal and lifestyle-driven expenditure shifts, often resulting in elevated false-positive rates during events such as holidays or one-time purchases.

5.3 Unsupervised Machine Learning Approaches

Given the absence of labeled data and the inherently imbalanced nature of personal financial records, unsupervised learning techniques dominate anomaly detection research in PFMS. Among these, Isolation Forest is frequently discussed due to its deviation-centric design. Rather than modeling normal behavior, this approach isolates anomalous observations through recursive partitioning, operating on the premise that anomalies are both rare and structurally distinct.

An anomaly score is computed based on the average path length required to isolate an observation across an ensemble of isolation trees, expressed as:

$$s(x, \psi) = 2^{-\frac{E(h(x))}{c(\psi)}} \tag{17}$$

where $E(h(x))$ represents the expected path length and $c(\psi)$ denotes the normalization factor derived from random binary tree properties. Higher scores indicate a greater likelihood of anomalous behavior. Comparative studies reported in the literature consistently demonstrate that Isolation Forest achieves superior detection capability relative to density-based and boundary-based alternatives, which often suffer from scalability limitations or sensitivity to parameter selection in PFMS environments.

5.4 Deep Learning-Based Anomaly Detection

Deep learning architectures are explored for anomaly detection in PFMS when spending behavior exhibits complex, non-linear patterns across high-dimensional feature spaces. Autoencoder-based models are commonly examined in this context, where the network learns a compressed representation of typical spending behavior and identifies anomalies based on elevated reconstruction error.

Despite their expressive power, deep learning approaches introduce practical challenges in PFMS deployment. These models typically require large volumes of high-quality training data and impose substantial computational overhead, which can limit real-time responsiveness in mobile or resource-constrained environments. Consequently, their adoption in mid-scale PFMS applications is often restricted to offline analysis or exploratory use rather than continuous monitoring.

5.5 Explainable Artificial Intelligence (XAI) Considerations

As PFMS transition from rule-based detection to learning-driven anomaly identification, issues of transparency and user trust become increasingly significant. Complex machine learning models often function as opaque systems, generating alerts without clear justification, which can reduce user confidence and adoption. In financial decision-support applications, the ability to explain detected anomalies is as critical as the detection itself.

To address this challenge, explainable artificial intelligence techniques are increasingly incorporated into PFMS pipelines. Post-hoc interpretability methods are applied to attribute anomaly decisions to specific spending features, enabling the generation of human-readable explanations. Additionally, attention-based mechanisms integrated within learning models highlight historical transactions that most strongly influenced anomaly detection outcomes.

The literature frequently advocates hybrid system designs in which unsupervised learning models perform anomaly detection, while rule-based or statistical logic is employed to generate interpretable explanations. Such hybrid implementations aim to balance analytical rigor with usability, ensuring that PFMS remain both technically robust and user-centric.

6 Group Expense Management and Collaborative Finance

6.1 Traditional Bill-Splitting Systems

Group expense management has traditionally been handled through manual and semi-digital methods, including spreadsheets, handwritten records, and basic ledger-based applications. In social and domestic contexts—such as shared travel, dining, or household utilities—tracking individual contributions accurately often becomes cumbersome. Early mobile applications for group finance typically function as electronic ledgers, requiring users to manually record transactions and assign participants for each expense. These systems commonly rely on static, rule-based logic, such as equal splitting, which fails to capture real-world variations where participants contribute unevenly. As a result, such approaches are prone to data entry errors, inconsistencies, and limited transparency, frequently leading to confusion and interpersonal friction within groups.

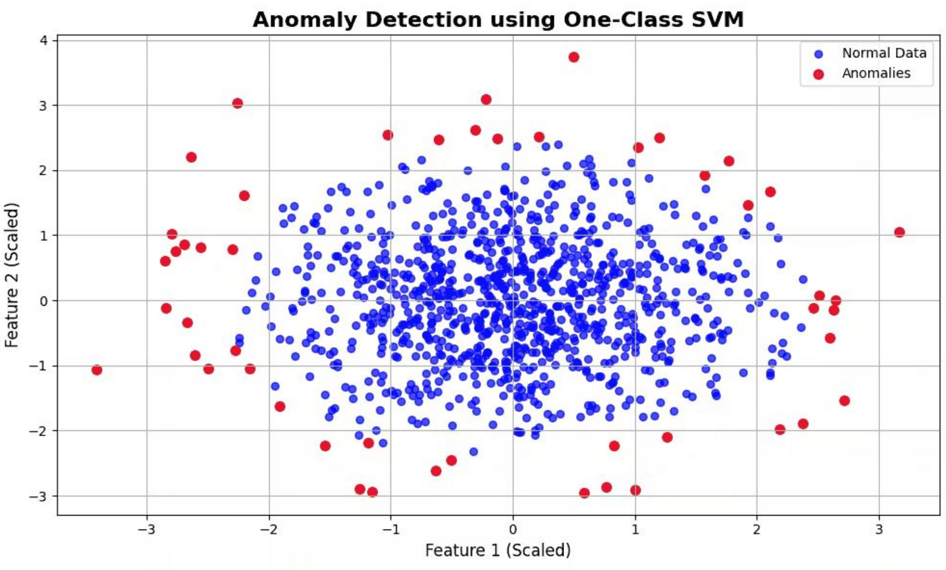

Figure 6: Anomaly detection using One-Class SVM

6.2 Graph-Based Settlement Optimization

To address the complexity inherent in multi-party debt resolution, contemporary group expense management systems incorporate graph-based settlement optimization techniques. In this formulation, group members are represented as nodes, while financial obligations are modeled as directed edges within a graph structure. Individual expenses are aggregated to compute net balances for each participant, and optimization algorithms are applied to minimize the total number of transactions required to settle group debts. Rather than executing numerous pairwise payments, the system recommends a reduced set of settlement transactions that efficiently balance obligations across the group. This approach improves clarity, reduces administrative overhead, and ensures consistency in large or frequently interacting groups.

6.3 Behavioral Analysis in Group Finance

Recent developments in collaborative finance emphasize behavioral and event-centric perspectives on shared expenses. Instead of treating expenses as isolated transactions, intelligent systems organize group spending around contextual events, such as trips, social gatherings, or recurring activities. This event-based organization enhances transparency by associating expenses, discussions, and settlements within a unified context. Dashboards summarizing group-level spending trends over monthly or annual periods enable participants to assess collective financial behavior and identify recurring patterns.

To further reduce manual effort, optical character recognition (OCR) technologies are increasingly integrated into group finance workflows. By automatically extracting item descriptions, prices, and totals from receipts, OCR-enabled systems streamline expense entry and reduce errors associated with manual transcription. This automation supports real-time collaboration while lowering the cognitive burden on users managing shared financial records.

6.4 Machine Learning–Based Payer Prediction and Expense Allocation

Advances in predictive analytics have enabled group expense management systems to evolve from passive tracking tools into proactive collaborative assistants. Machine learning techniques are applied to identify recurring behavioral patterns within groups and estimate the most likely payer for a newly recorded expense. In parallel, regression-based models are explored to recommend expense allocation ratios that reflect historical contribution behavior rather than uniform splits.

At a conceptual level, allocation logic can be expressed as a functional mapping between historical behavior and contextual attributes:

$$\text{Member\_Share} = f(\text{Historical Contribution}, \text{Group Persona}, \text{Event Type}) \tag{18}$$

By analyzing prior transactions and group dynamics, such systems can suggest personalized allocation strategies and notify users when projected spending approaches predefined group thresholds. The integration of classification for payer prediction and regression for allocation enhances coordination, reduces disputes, and supports more equitable financial collaboration within groups.

7 Comparative Analysis of Techniques

This section provides a qualitative synthesis of the methodological characteristics, data dependencies, and operational trade-offs associated with the techniques surveyed in this review. Rather than establishing hierarchical rankings, the analysis focuses on how architectural assumptions and modeling constraints align with distinct objectives in personal finance management. The discussion highlights interpretability, scalability, adaptability, and data requirements as central dimensions influencing methodological suitability.

7.1 Comparative Analysis of Budgeting Techniques

Budgeting mechanisms in Personal Finance Management Systems (PFMS) span a spectrum ranging from lightweight statistical heuristics to memory-driven deep learning architectures. The choice of budgeting technique is largely influenced by the balance between decision transparency and the capacity to capture complex behavioral patterns.

Table 1: Qualitative Comparison of Budgeting Techniques

Technique	**Data Requirements**	**Interpretability**	**Scalability**	**Adaptability**
EWMA	Low (minimal history)	High (transparent logic)	High (recursive updates)	Low (stable habits only)
Clustering (K-Means / GMM)	Medium (feature vectors)	Medium (cluster centroids)	High (unlabeled data)	Medium (pattern-based)
ARIMA	Medium (stationary series)	High (statistical formulation)	Low (category-wise modeling)	Medium (trend capture)
LSTM	High (long sequences)	Low (opaque representations)	Medium (resource intensive)	Very High (temporal shifts)

From a conceptual standpoint, budgeting approaches reflect a trade-off between temporal awareness and behavioral expressiveness. Statistical methods such as EWMA and ARIMA provide clear and interpretable mechanisms with low computational overhead, making them suitable for users whose spending behavior remains relatively stable over time. However, these approaches struggle to accommodate non-linear changes and seasonal variations. Clustering-based methods introduce a behavioral abstraction by grouping users or transactions into spending profiles, though they disregard transaction order and temporal evolution. In contrast, LSTM-based models excel at capturing long-range dependencies across multiple spending categories but require substantial historical data and computational resources, which may limit applicability for users with sparse financial records.

7.2 Comparative Analysis of Forecasting Techniques

Expenditure forecasting in PFMS involves balancing the stability of linear trend modeling with the flexibility required to represent non-linear and evolving financial dynamics.

Table 2: Qualitative Comparison of Forecasting Techniques

Technique	**Data Requirements**	**Interpretability**	**Scalability**	**Adaptability**
ARIMA / SARIMA	Low to Medium	High (linear formulation)	Low (parameter tuning)	Low (stationarity constraints)
LSTM / GRU	High (large datasets)	Low (hidden states)	Medium (computational overhead)	High (non-linear dynamics)
Hybrid (ARIMA–LSTM)	High (dual-stage modeling)	Medium (partial transparency)	Medium (integration complexity)	Very High (linear + non-linear)

The primary distinction among forecasting approaches lies in linear robustness versus non-linear flexibility. ARIMA-based models effectively capture autocorrelations and linear temporal dependencies, as expressed by:

$$Y_t = \phi_1 Y_{t-1} + \cdots + \theta_1 \epsilon_{t-1} + \epsilon_t \tag{19}$$

This formulation supports reliable short-term forecasting under stationary conditions but fails to represent complex behavioral shifts. Deep learning models, such as LSTM and GRU, address this limitation by modeling non-linear temporal relationships through memory mechanisms, albeit at the cost of increased computational demands and sensitivity to noise. Hybrid ARIMA–LSTM frameworks aim to reconcile these paradigms by extracting linear trends via ARIMA and modeling residual non-linear components using LSTM:

$$Y_{t,\text{ensemble}} = w_1 \cdot Y_{t,\text{ARIMA}} + w_2 \cdot h_{t,\text{LSTM}} \tag{20}$$

While such integration enhances forecasting robustness across heterogeneous financial contexts, it introduces additional architectural complexity and resource overhead.

7.3 Comparative Analysis of Anomaly Detection Techniques

Anomaly detection within PFMS is primarily formulated as an unsupervised learning problem, given the absence of labeled data defining undesirable or inefficient spending behavior.

The selection of an anomaly detection strategy reflects a balance between simplicity and expressive power. Rule-based systems provide transparent and verifiable baselines but lack the capacity to model multi-dimensional spending behavior or adapt to seasonal variation. Isolation Forest represents a paradigm shift by isolating anomalies directly through random partitioning rather than modeling normal behavior. Its anomaly scoring mechanism is defined as:

$$s(x, \psi) = 2^{-\frac{E(h(x))}{c(\psi)}} \tag{21}$$

Table 3: Qualitative Comparison of Anomaly Detection Techniques

Technique	**Data Requirements**	**Interpretability**	**Scalability**	**Adaptability**
Rule-Based	Low (manual rules)	Very High	High (low cost)	Very Low (rigid logic)
Isolation Forest	Low to Medium	Medium (requires XAI)	Very High ($O(n \log n)$)	High (feature-agnostic)
One-Class SVM	Medium	Low	Low (memory intensive)	Medium

This formulation enables efficient and scalable detection in large, unlabeled datasets, though it does not inherently provide explanatory context. More complex methods, such as One-Class SVM and Autoencoders, offer enhanced detection capabilities for intricate anomalies but suffer from poor scalability and high computational cost, particularly in mobile or real-time PFMS environments. As a result, contemporary systems often adopt hybrid designs, combining unsupervised detection mechanisms with rule-based or explainable AI layers to ensure interpretability and user trust.

8 Conclusion

The evolution of Personal Finance Management Systems (PFMS) reflects a broader shift from static, rule-based record-keeping tools toward intelligent, data-driven decision-support platforms. This transition is characterized by the convergence of four core analytical dimensions: adaptive budgeting, expenditure forecasting, anomaly detection, and collaborative group finance. Across the literature, budgeting mechanisms have progressed from fixed heuristics to behavior-aware approaches that incorporate temporal context and spending patterns to support personalized financial limits. Forecasting studies similarly demonstrate a movement toward hybrid modeling strategies, where statistical time-series methods are combined with learning-based approaches to improve robustness across diverse financial conditions.

Anomaly detection research within PFMS has increasingly favored unsupervised methodologies, motivated by the absence of labeled personal spending data and the highly individualized nature of financial behavior. Techniques designed to isolate deviations rather than classify predefined outcomes have enabled scalable identification of unusual spending patterns. In parallel, group expense management systems have advanced beyond manual tracking by incorporating automated settlement optimization, event-centric organization, and receipt digitization, thereby reducing administrative overhead and improving transparency in shared financial contexts.

Machine learning functions as a central enabler across these developments, facilitating the identification of complex patterns and long-term dependencies that are difficult to capture using traditional analytical methods. However, the growing reliance on complex models has also emphasized the importance of explainability, with explainable artificial intelligence emerging as a critical requirement for user trust and accountability in financial decision-support systems.

Despite substantial progress at the component level, existing research remains fragmented, with limited emphasis on unified system-level integration. The literature consistently indicates the need for cohesive intelligent PFMS frameworks that integrate budgeting, forecasting, anomaly detection, and collaborative finance within a single platform. Advancing toward such unified ecosystems represents a key direction for future research, with the potential to provide users with comprehensive financial insights, enhance financial literacy, and support long-term financial stability in an increasingly dynamic digital economy.

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